Root shape replay in the inherited-F4 oracle: a recorded negative

2026-10-15. `research/pdp_shape_replay_20261015/`. The protocol was registered (`d0b27a7f4`) before the claim ladders. **Class: engineering, a stage diagnostic.**

F4Remake's idea for the roots of the inherited-F4 decomposition oracle (Koblitz curves over $\mathbb{F}_{2^n}$, n prime). Rows that are dependent on earlier rows are recorded on the first root of a shape (layout, rows, degree, policy) and dropped from later roots of that shape. A replayed root spans a subspace, so the oracle stays exact and may split more. Recording is charged.

ladder	cells	default	replay	default ÷ replay	verdicts
frozen (pilot)	6	45,517,794	44,815,209	1.016	identical
`chain`	3	7,103,844	7,322,495	**0.970**	identical
`chain-holdout`	4	43,501,345	45,579,107	**0.954**	identical
`holdout`	2	41,414,967	42,137,329	**0.983**	identical

Replay costs more on all 9 registered cells (0.753–0.985×) and splits more on 6 of them. A counting build showed why. The large rank-deficient roots rarely repeat their layout across targets, so they are recorded but not reused: on `K_0/2^15`, $m = 4$, 18 roots were recorded and 6 replayed. Where roots do repeat, only about 4 rows are dropped: on `K_1/2^15`, $m = 4$, 1,832 replays dropped 3.8 rows each.

All predictions held: exact on 9/9 cells, every ratio in $[0.95, 1.10]$, and no gain. By the registered decision this lead is a **recorded null**: `KIC_F4_ROOT_REPLAY` stays off and the oracle is unchanged. The algebraic oracle remains far behind the root-finding table oracle ($4.1 \cdot 10^7$ word XORs against 592 multiplications per target at (23, 8)), which itself loses to rho, so no scoreboard row moves.

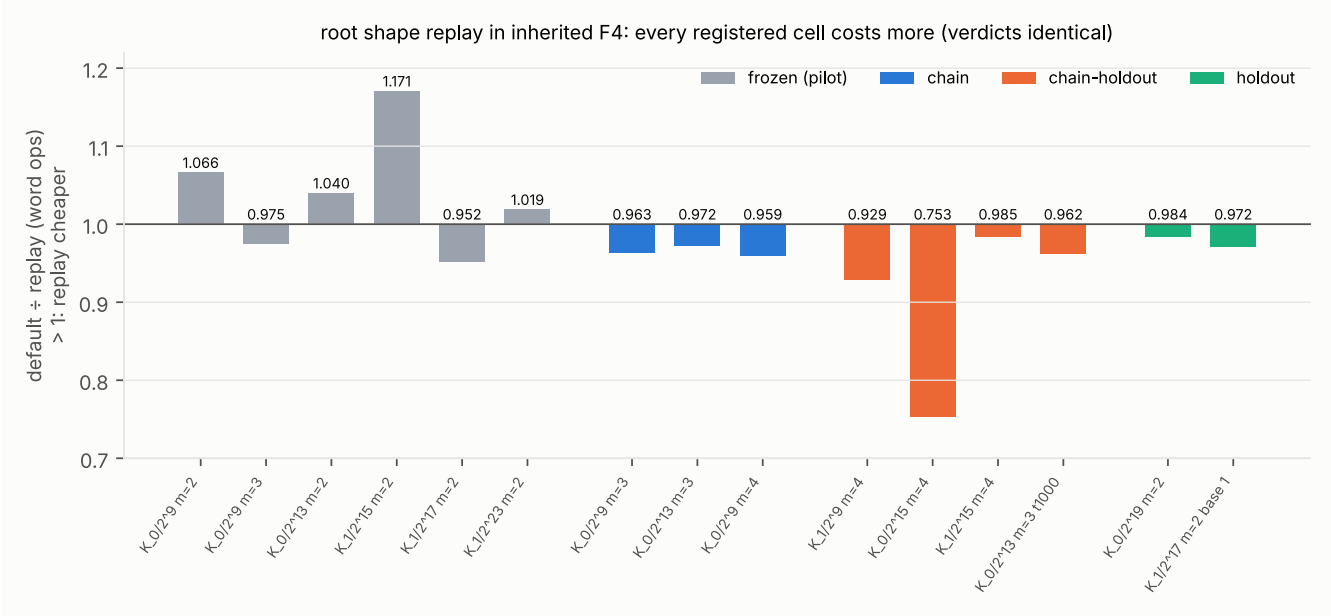


Figure 1: Default ÷ replay word operations per cell. Values above 1 mean replay is cheaper.

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU (shared). Counts are deterministic. Conductor task T-3.